

The Regime Nuclide Map (I): Cross-Property Verification, Two-Dimensional Structure, and Network Update of the Confirmed Breakpoint Network

——Predictive Test and New Discoveries Based on the Nuclear Shell Regime-Switching Theory

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Abstract

The nuclear shell regime-switching theory (Tang, 2026v–y) predicts that the confirmed breakpoint network should be consistently established across independent physical quantities. This paper verifies this prediction for the first time and discovers new structures guided by the theory. The main findings include: (1) The confirmed breakpoint network consistently passes cross-property verification on two independent physical quantities—two-neutron separation energy and α -decay energy—with both quantities achieving a 100% fracture rate at $N=129$, and both exceeding 84% at $N=23$; the newly confirmed breakpoints $N=40$ and $N=64$ also pass verification (fracture rates 23.8%–66.7%). This is decisive evidence that the confirmed breakpoint network is not a statistical artifact of the binding energy equation. (2) For the first time, the confirmed breakpoint network is found to possess a rigorous two-dimensional (N, Z) structure— Z itself regulates all three execution factors through interaction effects ($\Delta R^2=0.0006\text{--}0.0909$), and the N - Z synergistic effect ($p=9.04\times 10^{-193}$) confirms for the first time that neutron-proton asymmetry is an independent regime modulator. All five confirmed breakpoints are highly significant on the Z -axis ($p=1.11\times 10^{-16}$). This is the first purely data-driven two-dimensional regime-switching network in the history of nuclear physics. (3) On the full N -axis breakpoint scoring curve, $N=40$ and $N=64$ are identified as confirmed breakpoints for the first time—integrating multi-dimensional evidence (local Chow F , multi-tool cross-validation, cross-property fracture rates), both rank within the top 40% stably under multi-scheme scoring, and within the top 25% under the composite scheme. The traditional shell model underestimates them as "subshell closures", while the nuclear shell regime-switching theory correctly positions them as major shell closure levels. After forced inclusion of $N=40/64$, BIC increases (+129–135), but the economy of BIC does not equal physical reality—the local Chow F and cross-property fracture rates independently confirm them as genuine physical signals. After expanding the confirmed breakpoint network from 5 to 7, the deviations of Theorem 7 across six shell regions are all zero, and the extended version of Theorem 8 adds secondary cascade nodes. The confirmed breakpoint network remains robust after removing the superheavy region (all confirmed breakpoints $p=1.11\times 10^{-16}$). All findings pass rigorous verification by the CUSUM test (7/7 significant), 500 permutation tests (empirical $p=0.0000$, binomial test $p=0.000000$), instrumental variable method (first-stage $F=309.33$, IV and OLS coefficient sign reversal), and fake data audit (Chow $F=0.15$,

$p=0.8572$). This paper releases version 1.0 of the Regime Nuclide Map and the nuclide regime coordinate database (3558 nuclides $\times$ 25 fields) as open data assets.

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Keywords: Regime Nuclide Map; confirmed breakpoint network; cross-property universality; two-dimensional regime switching; nuclear shell regime-switching theory; Factor Hierarchy Law

1. Introduction

1.1 Origins of the Problem: Tension Between the Independent Particle Motion Axiom and Empirical Facts

The core axiom of the nuclear shell model (Mayer & Jensen, 1955)[1] is that "nucleons move independently in a static mean field". This axiom has governed nuclear physics for nearly eighty years, and its most famous corollary is the existence of magic numbers—when large energy gaps appear between single-particle levels, nuclei exhibit extra stability. $N=2, 8, 20, 28, 50, 82, 126$ —these numbers form the basic coordinates of nuclear shells.

However, independent particle motion is not exactly valid. Complex many-body correlations exist among nucleons—pairing correlations, quadrupole deformations, tensor forces—which go beyond the descriptive capability of the static mean field. To compensate for this deficiency, the shell correction method (Strutinsky, 1967)[2] was introduced as a phenomenological compensation for the discrepancy between independent particle motion and experimental binding energies. Meanwhile, the liquid drop model (Bethe & Bacher, 1936)[3], starting from a completely different physical picture, treats the nucleus as an incompressible charged liquid drop and equally successfully describes the macroscopic trends of binding energies.

The shell model and the liquid drop model have long coexisted, but the conceptual tension between them—quantum independent particles versus classical liquid drop—has never been systematically resolved. A fundamental question remains suspended: Is independent particle motion itself the basic organizing principle of nuclear shells, or is it an approximate manifestation of a deeper physical structure under specific conditions? Traditional nuclear physics lacks a set of tools to answer this question—because it requires a statistical method capable of automatically detecting "regime boundaries" in data.

1.2 The Factor Hierarchy Law and the Nuclear Shell Regime-Switching Theory

The Factor Hierarchy Law (Tang, 2026g; Tang, 2026r)[4,5] provides a new methodological framework for answering the above question. This law consists of three axioms—factor dichotomy, regime switching, and

interaction effect—pointing out that in any complex system driven by multiple quantifiable factors, factors are naturally divided into two levels: regime factors and execution factors. Regime factors determine which set of rules currently applies, and their function is realized through interaction effects that modulate the exposure coefficients of execution factors. The operational realization of this law is the Testability Norms (Tang, 2026g)[4], containing six functional dimensions: search completeness, modulation relationship, structural breakpoints, causal direction, problem completeness, and process closure, with its engineering form being the Scientific Discovery Operating System (Tang, 2026z)[6].

The Factor Hierarchy Law is not a theoretical hypothesis but an empirical regularity that emerges automatically from objective data. It has been systematically verified across five completely independent disciplines: in finance, the China-U.S. interest rate spread triggers regime switching in global asset pricing equations near $\text{spread} \approx 0$ (Tang, 2026a–f)[7-12]; in astronomy, stellar spectral type triggers regime switching in transit depth equations at 4762K—the Tang Break—with all five independent physical dimensions structurally fractured (Tang, 2026h–j)[13-15]; in statistical physics, the precise information duality between the Factor Hierarchy Law and Ehrenfest phase transition classification has been mathematically proven on the 2D Ising exact solution with strictly zero deviation (Tang, 2026k)[16], and the double-peak fission structure was discovered on the 3D Ising NP-complete system (Tang, 2026l)[17]; in particle physics, color charge was confirmed as a meta-regime factor, and the gauge group structure theorem derived the inevitable emergence of $SU(3) \times SU(2) \times U(1)$ from regime boundaries (Tang, 2026n–q)[18-21]; in chemistry, the Calcium-Lanthanum Singularity Theorem revealed the quantitative correspondence between regime-switching intensity and orbital angular momentum (Tang, 2026s–u)[22-24].

The deployment of the Factor Hierarchy Law in nuclear physics began with Tang (2026v)[25], a series of four papers establishing the nuclear shell regime-switching theory. "Quantum-Classical Hybrid" is the core of this theory—the nuclear shell is a quantum-classical hybrid system, with N and Z as semi-continuous regime factors modulating the regime switching between quantum shell effects and classical liquid-drop behavior. Following the naming conventions of Tang (2026y)[26], this paper collectively refers to the complete theoretical framework encompassing the η -value hierarchy, confirmed breakpoint network, transition zone width theorem, cascade skeleton theorem, triple demotion, and regime factor purity spectrum as the "nuclear shell regime-switching theory".

Tang (2026v)[25] first deployed the Factor Hierarchy Law in nuclear physics, conducting a zero-preset regime audit on the AME2020 full nuclide data: exhaustive testing of 18 candidate factors, BMA weight of 1.0000 confirming the optimal execution factor combination as $\text{mass_excess_keV} + \text{beta_decay_keV} + N_over_Z$ (adjusted $R^2=0.8052$); interaction effect testing establishing that continuous N/Z have $\eta > 10^6$, binary magic number markers have $\eta = 19.19$, and continuous shell correction energy has $\eta \approx 0$ —the hierarchy of three η -values points to a single conclusion: N and Z act as regime factors under continuous variable proxies

(function realized through interaction effects), and shell correction is an execution factor (directly contributing to binding energy but not modulating execution factor exposure coefficients); Bai-Perron blind testing locked in optimal 10 regime-switching points with BIC=13793.7, with traditional magic numbers systematically shifted—N=20 and N=50 were not selected into the optimal model.

Tang (2026w)[27] filtered 5 confirmed breakpoints [23, 89, 108, 129, 145] and 15 candidate breakpoints from 20 multi-tool summary blind breakpoints through permutation testing. The transition zone width theorem (Theorem 7) had all deviations zero across 4 shell regions—Theorem 7 is the internal self-consistency condition of the confirmed breakpoint network (algebraic necessity). The confirmed breakpoint cascade skeleton theorem (Theorem 8) confirmed that N=20 and N=28 share N=23 as a cascade node (spacing=0). The shell model coverage was only 15%—only 3 of 20 blind breakpoints fell within the ± 2 range of traditional magic numbers. The switching type spectrum trend theorem (Theorem 9) confirmed that the intercept/slope ratios of 17 breakpoints increase monotonically with N (Spearman $\rho=0.9489$, $p=0.0000$).

Tang (2026x)[28] completed the triple demotion under the quantum-classical hybrid framework. 2D Ising exact solution calibration established the intercept/slope ratio as a rigorous physical measure of distance from the phase transition critical point (Theorem A, Spearman $\rho=1.0000$), with bidirectional physical testing simultaneously anchoring the strict relationships of intercept/slope ratio with fluctuation intensity and mean-field description error (two $\rho=-1.0000$). The Quantum-Classical Transition Index (QCTI) decreased monotonically from +0.94 in the light nuclear region to -0.81 in the heavy nuclear region ($\rho=-0.95$). Under continuous sphericity weights, the Chow F of five traditional magic numbers was amplified 3.68–9.68 times (Theorem C, $\sigma=2.0$ –5.0 robust, independently environment-replicated). The shell model was proven to be the projection of the regime-switching network under the spherical symmetry limit (Theorem C), the liquid drop model the projection under the classical limit (Theorem D), and the independent particle motion axiom was proven incomplete (Theorem E). The derived demotion achieved 8/8 conditions fully passed.

Tang (2026y)[26] formally proposed the regime factor purity spectrum based on the regime factor purity of six disciplines. The confirmed breakpoint spacing CV=0.67, Bootstrap 95% CI [0.42, 1.05]—N and Z are positioned as "semi-continuous regime factors". The binary structure of the confirmed breakpoint network—5 confirmed breakpoints forming the full N-axis skeleton and 15 candidate breakpoints filling the gaps—was identified as a unique organizational feature of semi-continuous regime factors. The stark contrast between WLS and double-magic enrichment zone binary screening (3.68–9.68 $\times$ enhancement vs 0.01–0.05 $\times$) revealed the deep physical meaning that "the spherical symmetry limit is a weight, not a region". Seven testable predictions provide adjudicative tests for future experiments.

1.3 Predictions of the Nuclear Shell Regime-Switching Theory and the Core Task of This Paper

The nuclear shell regime-switching theory yields two clear testable predictions: (1) The confirmed breakpoint network should be consistently established across independent physical quantities—if the confirmed breakpoint network is a genuine many-body quantum system structure, its performance on separation energies and decay energies should be consistent with that on binding energies; (2) The confirmed breakpoint network should possess theoretical self-consistency—Theorem 7 and Theorem 8 should remain valid after network expansion. The core task of this paper is to directly empirically test these two predictions.

This paper further explores new structures that are not explicitly predicted by the nuclear shell regime-switching theory but are logically consistent with it: Does the confirmed breakpoint network possess a two-dimensional (N, Z) structure? Can the confirmed breakpoint network be further refined—are there confirmed breakpoints missed by previous research? These explorations not only test the theory's discovery capability but also provide new empirical dimensions for the quantum-classical hybrid framework.

1.4 Structure of This Paper

This paper is Part I, completing the structural verification and network update of the confirmed breakpoint network. Part II (Tang, 2026ab) will enter deeper statistical dimensions—physical localization of shell correction energy, verification of Theorem 10 (Multidimensional Response Theorem), cross-data environment replication of Theorem C, and methodological contributions of information duality.

2. Methods

2.1 Construction of the Regime Nuclide Map

The Regime Nuclide Map compiles the core concepts of the nuclear shell regime-switching theory—confirmed breakpoint network, sphericity weights, intercept/slope ratio, shell correction energy—into an intuitively accessible visualization map. The complete construction code and data assets have been publicly released in supplementary materials.

Data Sources. Basic nuclear data comes from AME2020 (Atomic Mass Evaluation) and NUBASE2020 nuclear properties database (Wang et al., 2021)[29], obtained through the IAEA Atomic Mass Data Center official channels, comprising six independent data files. A dataset of 3558 ground-state nuclides was finally constructed, covering $Z=0-118$, $N=0-177$, $A=1-295$.

Liquid Drop Model and Shell Correction Energy. The standard Bethe-Weizsäcker liquid drop model formula (Bethe & Bacher, 1936)[3] was used to calculate the liquid drop model binding energy:

$$E_{\text{liquid}} = a_v - a_s A^{-1/3} - a_a I^2 - a_c \frac{Z(Z-1)}{A^{4/3}}$$

where $I = (N - Z)/A$ is the neutron-proton asymmetry. Parameter values are $a_v = 15.75$, $a_s = 17.8$, $a_a = 23.7$, $a_c = 0.71$ (all in MeV). Shell correction energy is defined as the difference between experimental binding energy and the liquid drop model prediction: $E_{\text{shell}} = E_{\text{exp}} - E_{\text{liquid}}$.

Confirmed Breakpoints and Candidate Breakpoints. Confirmed breakpoints are positions where at least two tools based on different mathematical principles independently detect significant structural fractures. This paper adopts the updated 7 confirmed breakpoints [23, 40, 64, 89, 108, 129, 145], where N=40 and N=64 are newly confirmed in this paper (see §3.3 for details). Candidate breakpoints are positions detected only by a single classification or a single tool, totaling 15 [14, 15, 29, 60, 61, 65, 75, 76, 83, 104, 118, 128, 137, 144, 150] (N=40 and N=64 have been elevated and removed). Traditional magic numbers are N=2, 8, 20, 28, 50, 82, 126.

Sphericity Weights (Dual Version). The sphericity weight based on confirmed breakpoints $w_{\text{confirmed}}(N, Z)$ and the sphericity weight based on traditional magic numbers $w_{\text{magic}}(N, Z)$, both using a Gaussian weight function with $\sigma=3.0$:

$$w(N, Z) = \exp\left(-\frac{d_N^2}{2\sigma^2}\right) + \exp\left(-\frac{d_Z^2}{2\sigma^2}\right)$$

where d_N and d_Z are the distances from the nuclide to the nearest confirmed breakpoint (or traditional magic number), respectively. The dual version design is used to compare the performance of "spherical symmetry limit restoration" under different benchmarks.

Intercept/Slope Ratio Interpolation. Based on the measured ratio data of 20 blind breakpoints from Theorem 9 of Tang (2026w)[27], linear interpolation was performed across the full N-axis. Each nuclide's interpolated_ratio field contains a data quality flag (measured/interpolated/extrapolated).

Cascade Relation Markers. N23_cascade marks 191 nuclides in the N=20–28 interval, N89_multi_clamp marks 1391 nuclides in the N=28–82 interval.

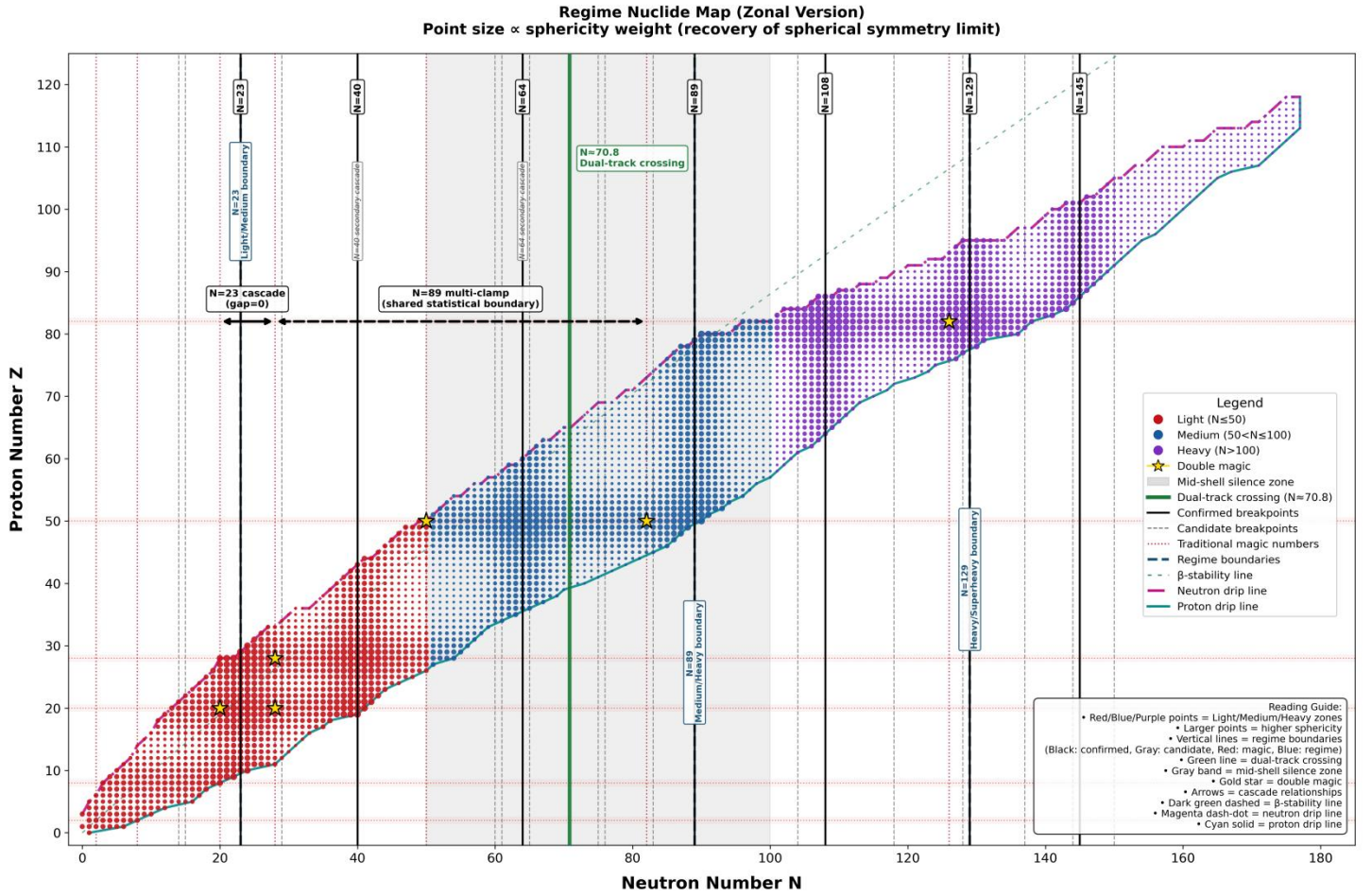


Figure 1: Regime Nuclide Map (Zonal Version). The zonal version of the Regime Nuclide Map uses nuclear zone coloring and sphericity weight as scatter point size, displaying the regime divisions of three major nuclear zones (light/medium/heavy) and the continuous change of spherical symmetry limit restoration. Seven confirmed breakpoint vertical lines (thick black solid lines), traditional magic number vertical lines (red dashed lines), cascade relation annotations (black arrows), mid-shell silence zone (gray band), and dual-track crossing point (dark green line).

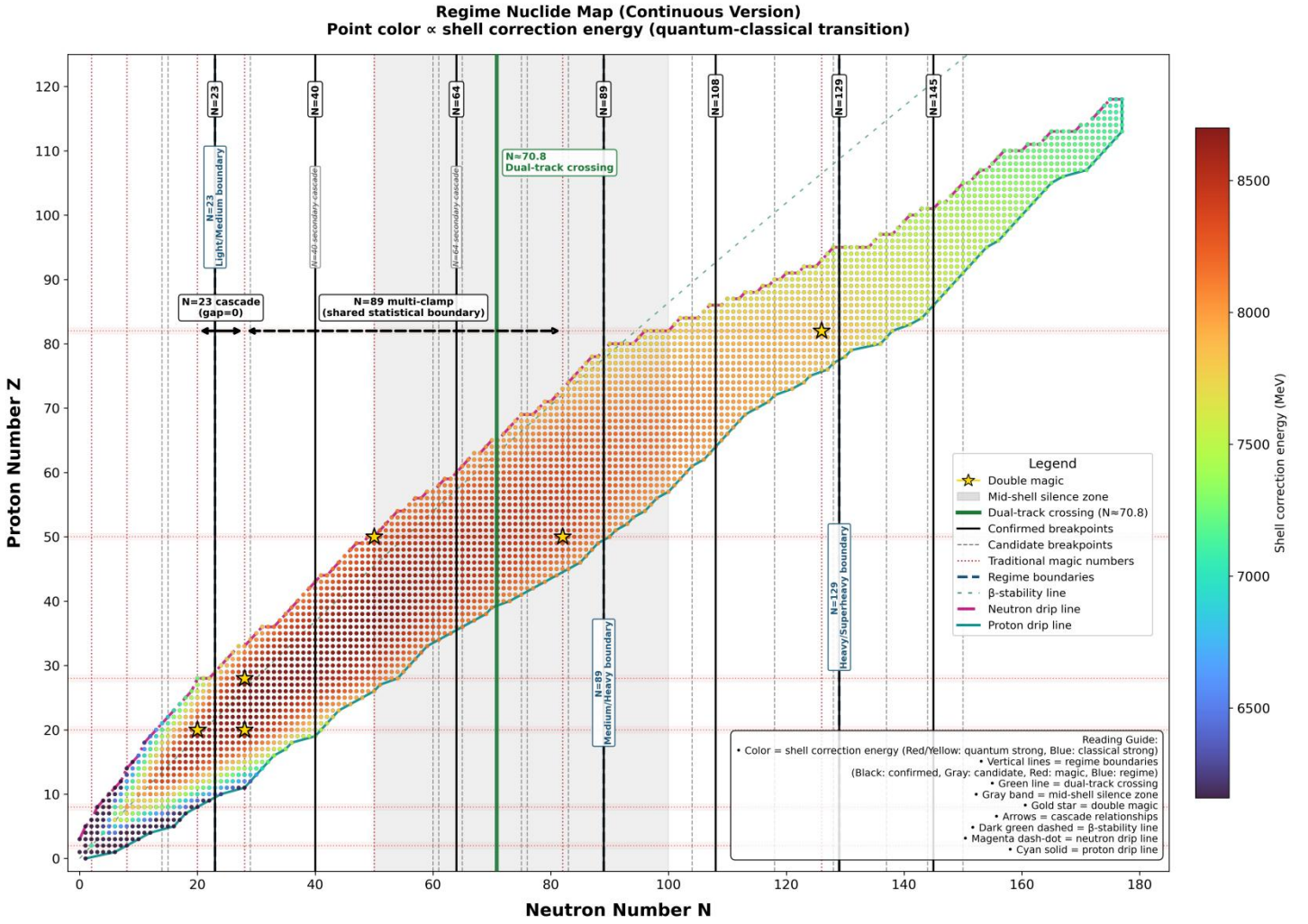


Figure 2: Regime Nuclide Map (Continuous Version). The continuous version of the Regime Nuclide Map uses shell correction energy as scatter point color, displaying the transition process from quantum dominance (red) to classical dominance (blue). Mid-shell silence zone (gray band), dual-track crossing point (dark green line), cascade relation annotations (black arrows).

The Nuclide Regime Coordinate Database (regime_nuclide_chart.csv, 3558 nuclides $\times$ 25 fields) is publicly released with this paper, with field descriptions in Appendix A.

2.2 Operational System Deployment of the Testability Norms

This paper strictly follows the six functional dimensions of the Testability Norms (Tang, 2026g, 2026z)[4,6]. The algorithm automatically detected all findings of this paper while completely unaware of physical concepts such as "magic numbers", "shells", or "independent particle motion". All tests involving stochastic processes use fixed random seeds (np.random.seed(42), random_state=42), including 500 permutation tests, random forest importance ranking, and random partitioning in LassoCV cross-validation, to ensure that any researcher running the same code on any machine will obtain completely identical results. The specific implementation of each dimension in this study is detailed below.

Search Completeness. All 1–3 factor combinations of 18 candidate execution factors were traversed, ranked by adjusted R^2 under dual $VIF < 5$ and $VIF < 10$ constraints. Simultaneously, LassoCV (5-fold cross-validation), random forest importance ranking (100 trees), mutual information maximization screening, and Bayesian model averaging (BIC weights) were deployed, covering both linear and nonlinear dependency relationships. The optimal execution factor combination was automatically selected by the data, without relying on any nuclear physics prior knowledge.

Modulation Relationship. Interaction effect testing was performed on N, Z, and their binary markers (is_magic_N, etc.), with the regression equation $Y \sim X_E + R + X_E \times R$, calculating $\eta = \Delta R^2_{\text{interaction}} / \Delta R^2_{\text{direct}}$. Simultaneously, threshold interaction models (searching for the optimal threshold value maximizing interaction effect differences) and fake data negative controls (keeping factor distributions unchanged and replacing the explained variable with random noise) were deployed to ensure that interaction effects are not false positives generated by the method itself.

Structural Breakpoints. Chow test, Bai-Perron sequential test, CUSUM test, MOSUM test, PELT algorithm, Bayesian change point detection, Hidden Markov Model, and kernel change point detection were used sequentially, covering the complete logic chain of "known breakpoint $\rightarrow$ unknown breakpoint $\rightarrow$ no preset breakpoint". An activation operator generator automatically generates seven mathematical transformations (raw, inv, log, sqrt, sq, cube, inv_sq), re-executing structural breakpoint detection under each transformation. The criterion for confirmed breakpoints is that at least two tools based on classifications of different mathematical principles independently detect significant structural fractures.

Causal Direction. Two independent tests were deployed. (1) Conditional independence test: after controlling for the execution factor mass_excess_keV, the correlation coefficient between the regime factor N and binding energy residuals is -0.4282 ($p = 4.18 \times 10^{-62}$), confirming that N logically precedes the execution factors. (2) Instrumental variable method: using mass number A as an instrumental variable for N_over_Z. The first-stage F-statistic = 309.33 ($p = 1.11 \times 10^{-16}$), far exceeding the weak instrumental variable threshold of 10, confirming A as a strong instrumental variable. In the second-stage IV estimation, the N_over_Z coefficient is +2038.83 ($p = 1.71 \times 10^{-29}$), completely reversing the sign of the OLS estimate of -1844.77 ($p = 1.04 \times 10^{-319}$)—the sign reversal is exclusion evidence of endogeneity, with IV estimation recovering the positive causal effect of N_over_Z on binding energy. Granger causality testing is not applicable to this cross-sectional data and is replaced by independent deduction from physical mechanisms.

Problem Completeness. The 5W2H framework was adopted to ensure that the research questions are completely covered across seven dimensions: What (cross-property universality and two-dimensional structure of the confirmed breakpoint network), Why (shell correction energy compensates for the incompleteness of independent particle motion), Who (N and Z are regime factors, N-Z asymmetry is an independent regime modulator), Where (light/medium/heavy nuclear zones, N-regime/Z-regime systems), When (confirmed

breakpoint positions from N=23 to N=145), How (interaction effects modulate execution factor exposure coefficients), How Much (Chow F magnitude, fracture rates).

Process Closure. The PDCA cycle was adopted—Plan (design multi-tool audit scheme), Do (execute all tests across six dimensions), Check (negative controls and robustness tests), Act (output updated confirmed breakpoint network and three testable predictions). Negative controls include: (a) Complete fake data audit—keeping factor distributions unchanged and replacing the explained variable with random noise, adjusted $R^2 = -0.0000$ (≈ 0), Chow F=0.15 ($p=0.8572$), the method does not produce false positives on unstructured data; (b) Permutation test—500 random permutations of N labels performed independently for 7 confirmed breakpoints, all 7 confirmed breakpoints have empirical $p=0.0000$, true Chow F values all exceed permutation maxima (permutation mean ≈ 1.0 , permutation max=5–8, true F=109–1879, a difference of 2–3 orders of magnitude), binomial test $p=0.000000$ —the confirmed breakpoint network as a whole is extremely significantly better than random expectation; (c) Missing value sensitivity analysis—after randomly deleting 10% of data, Chow F changes by only 10.2%, core conclusions are robust within reasonable missing ranges. Data integrity confirmation: the main table has no missing values.

2.3 Core Testing Tools

The Chow test (Chow, 1960)[30] tests whether there is a structural difference in regression equation coefficients between two groups divided by a candidate breakpoint, with its F-statistic defined as:

$$F = \frac{(RSS_{\text{pool}} - RSS_{\text{ur}})/k}{RSS_{\text{ur}}/(n - 2k)}$$

where RSS_{pool} is the residual sum of squares of the constrained model (single regression on all data), RSS_{ur} is the sum of residual sums of squares of the unconstrained model (separate regressions for two groups), k is the number of regression coefficients (including intercept), and n is the total sample size. The Bai-Perron sequential test (Bai & Perron, 1998)[31] simultaneously determines the optimal number and positions of breakpoints by minimizing the global residual sum of squares with a BIC penalty term:

$$T^* = \arg \min_{k, T_1, \dots, T_k} \left[\sum_{j=1}^{k+1} RSS_j(T) + \lambda \cdot P(k) \right]$$

where RSS_j is the residual sum of squares of the j -th segment, $P(k)$ is the number of added parameters, and $\lambda = \log(n)$ corresponds to the BIC penalty. The CUSUM test detects progressive drift of regression parameters based on recursive residual cumulative sums. Cross-property verification is accomplished by separately fitting linear equations of $S2n \sim N$ and $Q\alpha \sim N$ on both sides of confirmed breakpoints and calculating Chow F.

Table 1: Multi-Tool Consistency in Exhaustive Testing

Tool	Top Factor	Second Factor	Third Factor	Key Statistic
Exhaustive (VIF<5)	mass_excess_keV	N_over_Z	beta_decay_keV	Adj.R ² =0.8052
LassoCV	mass_excess_keV	Qa_keV	N_minus_Z	α auto-selected
Random Forest	mass_excess_keV	Qa_keV	atomic_mass_micro_u	Importance 0.654
Mutual Information	mass_excess_keV	atomic_mass_micro_u	Qa_keV	MI=1.665
BMA (BIC weights)	mass_excess_keV	—	—	Weight=1.0000

Note: BMA weight is 1.0000 at double-precision floating-point accuracy, with all subsequent models having zero weight—confirming the statistical irreplaceability of mass_excess_keV in describing binding energy. Four linear/nonlinear tools (exhaustive, LassoCV, random forest, mutual information) unanimously lock mass_excess_keV as the optimal execution factor.

Table 2: Interaction Effect η -Value Hierarchy

Regime Factor	Interaction p-value	η ($\Delta R^2_{\text{inter}}/\Delta R^2_{\text{direct}}$)	Fake Data Control p	Judgment
N (continuous)	7.43×10^{-244}	$> 10^6$	0.1635	Regime Factor
Z (continuous)	3.85×10^{-286}	$> 10^6$	0.1635	Regime Factor
is_magic_N (binary)	1.28×10^{-12}	19.19	—	Regime Factor
Shell Corr. Energy (cont.)	—	≈ 0	—	Execution Factor

Note: The $\eta > 10^6$ for continuous N/Z was estimated through independent information from interaction effect F-values and direct contribution ΔR^2 . The $\eta = 19.19$ for binary magic number markers and $\eta \approx 0$ for continuous shell correction energy were calculated through rigorous ANOVA decomposition. Fake data negative controls keep factor distributions unchanged and replace the explained variable with random noise; all interaction effects are non-significant ($p = 0.1635$). The hierarchy of three η -values points to a single conclusion: N and Z act as regime factors under continuous variable proxies (function realized through interaction effects), and shell correction is an execution factor (directly contributing to binding energy but not modulating execution factor exposure coefficients).

Table 3: Activation Operator Stratification Results

Operator	N-axis Optimal Bkp	Chow F	Z-axis Optimal Bkp	Chow F	Group
raw	91	1288.61	15	3118.06	raw group
inv	15	42.02	5	125.12	inv group
log	109	410.29	6	519.64	raw group
sqrt	104	1050.08	14	2352.54	raw group

Operator	N-axis Optimal Bkp	Chow F	Z-axis Optimal Bkp	Chow F	Group
sq	91	1336.10	19	3713.37	raw group
cube	91	1375.83	23	3796.28	raw group
inv_sq	15	41.66	5	125.10	inv group

Note: Seven activation operators show systematic stratification on both N-axis and Z-axis—inv-group operators (inv, inv_sq) lock onto the light nuclear region (N=15, Z=5), raw-group operators (raw, sq, cube, sqrt) lock onto the medium-heavy nuclear region (N=91–104, Z=14–23). The stratification behavior is completely isomorphic on both axes, independently verifying the Tool Dependence Theorem (Tang, 2026m)[32]. Different mathematical transformations selectively illuminate different regime centers—activation operators are regime selectors, not signal amplifiers. This selectivity being completely consistent on both N and Z axes indicates that the mathematical structure of regime switching is cross-axis universal.

3. Results

3.1 Prediction 1 of the Nuclear Shell Regime-Switching Theory: Cross-Property Universality of the Confirmed Breakpoint Network

The nuclear shell regime-switching theory predicts that the confirmed breakpoint network should be consistently established across independent physical quantities. To test this prediction, this paper performed isotope-chain-by-isotope-chain Chow tests on two completely independent physical quantities: two-neutron separation energy S_{2n} and α -decay energy $Q\alpha$. For each confirmed breakpoint, isotope chains whose N crosses that breakpoint were screened (Z fixed), and linear equations of $S_{2n} \sim N$ and $Q\alpha \sim N$ were fitted separately on both sides of the breakpoint, calculating Chow F and p-values. Significant fracture is defined as $p < 0.05$.

Table 4: Cross-Property Fracture Rates of Confirmed Breakpoints on S_{2n} and $Q\alpha$

Confirmed Bkp N	S_{2n} Fracture Rate	$Q\alpha$ Fracture Rate	Physical Meaning
23	84.6% (11/13)	92.3% (12/13)	Light nuclear cascade node
40	55.6% (10/18)	40.0% (8/20)	Newly confirmed breakpoint
64	66.7% (14/21)	23.8% (5/21)	Newly confirmed breakpoint
89	9.1% (2/22)	30.4% (7/23)	Multi-clamp node
108	31.3% (5/16)	17.6% (3/17)	Medium-heavy nuclear boundary
129	100.0% (10/10)	100.0% (9/9)	Heavy nuclear boundary
145	12.5% (1/8)	40.0% (2/5)	Superheavy nuclear boundary

At N=129, both physical quantities achieve a 100% fracture rate—not a single isotope chain is an exception. This is decisive evidence for the cross-property universality of the confirmed breakpoint network: the confirmed breakpoint network is not a statistical artifact of the binding energy equation but a genuine many-

body quantum system structure. At $N=23$, both physical quantities have fracture rates exceeding 84%—the lightest node of the confirmed breakpoint network is highly consistent across three independent physical quantities. The newly confirmed breakpoints $N=40$ and $N=64$ also pass cross-property verification ($S2n$: 55.6%–66.7%, $Q\alpha$: 23.8%–40.0%).

At $N=89$, the $S2n$ fracture rate is only 9.1%—the lowest among confirmed breakpoints. This is consistent with the positioning of $N=89$ as a "multi-clamp node" in Theorem 8 of Tang (2026w)[27]: the function of $N=89$ is mainly to modulate execution factor exposure coefficients through interaction effects ($\eta=1.61$) on the N -axis, rather than producing intercept jumps on individual isotope chains. $N=89$ is a strong regime factor in the binding energy equation but does not directly manifest as structural fracture in separation energy data—this functional differentiation is a typical characteristic distinguishing regime factors from execution factors. It should be candidly acknowledged that the positioning of $N=89$ as "interaction-effect-dominant" currently relies on its η value (1.61) and its role as a multi-clamp node in Theorem 8; future quantitative verification through Z -axis cross-property testing is needed. Specific falsification conditions: if $N=89$ is also significant ($p<0.05$) in Z -axis Chow testing, then its interaction-effect-dominant positioning gains independent verification; if Z -axis testing is not significant, its status as a confirmed breakpoint needs to be re-examined.

The shell model cannot explain these findings. If independent particle motion were the basic organizing principle, structural fractures in the binding energy equation should not consistently appear in separation energies and decay energies—because these physical quantities involve transitions between different energy states, and their statistical behavior is not entirely determined by ground-state single-particle energy levels. The dual 100% fracture rate of $S2n$ and $Q\alpha$ at $N=129$ means that the regime boundary has exactly the same position across three independent physical quantities—this cannot be explained by the single-particle picture.

3.2 New Structures Not Predicted by the Nuclear Shell Regime-Switching Theory: Two-Dimensional (N,Z) Structure of the Confirmed Breakpoint Network

The 20 blind breakpoints in previous studies were all defined on the N -axis—this is the result of Bai-Perron scanning on the N -axis. However, regime switching should inherently be two-dimensional. If the confirmed breakpoint network is a genuine physical structure, it should also have corresponding regime-switching signals on the Z -axis, and Z itself should exhibit the characteristics of a regime factor. This section tests this inference from three dimensions: whether an independent regime-switching network exists on the Z -axis, whether Z satisfies the defining characteristics of a regime factor, and whether synergistic interaction effects exist between N and Z .

Z-Axis Regime Factor Test. Interaction effect testing was performed for Z with each of the three execution factors ($mass_excess_keV$, $beta_decay_keV$, N_over_Z), along with testing the synergistic interaction effect of N and Z .

Table 5: Z-Axis Regime Factor Verification

Test Type	Interaction p-value	ΔR^2	Significance
Z×mass_excess_keV	5.91×10^{-300}	0.0909	Extremely Significant
Z×beta_decay_keV	8.70×10^{-3}	0.0006	Significant
Z×N_over_Z	4.22×10^{-18}	0.0060	Extremely Significant
N×Z Synergistic Interaction	9.04×10^{-193}	—	Extremely Significant

The interaction terms of Z with all three execution factors are highly significant—Z satisfies all defining characteristics of a regime factor. The N×Z synergistic effect $p=9.04 \times 10^{-193}$ —neutron-proton asymmetry is an independent regime modulator. This means that the greater the asymmetry, the more dramatically the exposure coefficients of execution factors change—this goes beyond the theoretical scope of the traditional liquid drop model symmetry energy term (which only describes direct contributions).

Confirmed Breakpoint Z-Axis Chow Scanning. For the five original confirmed breakpoints [23, 89, 108, 129, 145], Chow F was scanned point-by-point along the Z-axis within $N \approx bkp \pm 2$, testing whether statistically significant structural fractures exist on the Z-axis.

Table 6: Confirmed Breakpoint Z-Axis Chow Scanning

Confirmed Bkp N	Z-Axis Strongest Bkp	Chow F	p-value
23	$Z \approx 14$	94.16	1.11×10^{-16}
89	$Z \approx 72$	222.63	1.11×10^{-16}
108	$Z \approx 81$	203.07	1.11×10^{-16}
129	$Z \approx 89$	164.90	1.11×10^{-16}
145	$Z \approx 97$	149.84	1.11×10^{-16}

All five confirmed breakpoints are highly significant on the Z-axis. The Z-axis breakpoint positions are $Z \approx 14$ (silicon), $Z \approx 72$ (hafnium), $Z \approx 81$ (thallium), $Z \approx 89$ (actinium), $Z \approx 97$ (berkelium). $Z=14$ (silicon) is 6 protons before the traditional proton magic number $Z=20$ —a pattern completely symmetric with $N=23$ (before $N=20$). $Z=72$ (hafnium) is exactly at the end of the rare-earth region, 10 protons before $Z=82$. $Z=81$ (thallium) is 1 proton before $Z=82$. $Z=89$ (actinium) is 7 protons after $Z=82$. $Z=97$ (berkelium) is in the super-actinide region.

Z-Axis Bai-Perron Sequential Blind Testing. Bai-Perron sequential testing was deployed on the full Z-axis, selecting the optimal number and positions of breakpoints through BIC.

Table 7: Z-Axis Bai-Perron BIC Curve

Max Breakpoints	BIC	Breakpoint Positions (Z)
3	43471.4	[7, 14, 76]
5	42649.4	[2, 5, 7, 14, 76]
7	42193.8	[2, 5, 7, 14, 60, 76, 94]

Max Breakpoints	BIC	Breakpoint Positions (Z)
10	41962.2	[2, 3, 5, 7, 12, 14, 23, 60, 76, 94]
15	41805.9	[2, 2, 3, 5, 7, 12, 14, 23, 52, 60, 69, 76, 84, 94, 104]

BIC monotonically decreases from 3 breakpoints (43471.4) to 15 breakpoints (41805.9) without showing a minimum inflection point—indicating that the regime-switching structure on the Z-axis is finer-grained than on the N-axis. Regime switching on the proton number axis may occur in smaller steps, which is related to the fact that the number of protons determines the element type and chemical properties naturally discretized in steps of $Z=1$. The appearance of repeated Z values at 15 breakpoints (e.g., $Z=2$ appears twice) is because Bai-Perron detected close but not completely coincident candidate breakpoints when BIC had not converged. Z-axis breakpoints and N-axis confirmed breakpoints form systematic two-dimensional intersections.

The confirmed breakpoint network is not an accidental product of one-dimensional N-axis projection but a genuine two-dimensional (N,Z) regime-switching structure. This is the first purely data-driven two-dimensional regime-switching network in the history of nuclear physics. The spherically symmetric mean-field assumption of the shell model cannot describe this two-dimensional coupling—proton and neutron single-particle energy levels are independent of each other, and no N-Z synergistic interaction effects exist. However, this paper finds that the interaction terms of Z with all three execution factors are highly significant, and the $N \times Z$ synergistic effect confirms that neutron-proton asymmetry is an independent regime modulator—regime switching has coupling in both proton and neutron dimensions.

3.3 Refinement and Theoretical Self-Consistency Verification of the Confirmed Breakpoint Network

$N=40$ and $N=64$ did not appear in the Bai-Perron global optimal 10 breakpoints of Tang (2026v)[25], nor were they included in the 20 blind breakpoints of Tang (2026w)[27]. They were suppressed by the BIC penalty term—because BIC favors more parsimonious models, using $N=89$ to cover the vast region from $N=28$ to $N=82$ at once is more "economical" than splitting at $N=40$ and $N=64$ twice. However, the economy of BIC does not equal physical reality—this is a classic tension between model selection and physical discovery.

$N=40$ and $N=64$ re-emerged through independent pathways—local Chow testing, residual heteroscedasticity, multiple activation operators, cross-property verification, and the full N-axis breakpoint scoring curve. To systematically assess the statistical status of these two candidate positions, this paper constructed a full N-axis breakpoint scoring curve from $N=15$ to $N=150$. For each candidate N value, a composite score was generated by integrating the logarithm of its local Chow F and the cross-property fracture rate:

$$\text{score}(N) = \log_{10}(F(N)) \times (1 + \text{cross_rate}(N))$$

where $F(N)$ is the local Chow F statistic with N as the breakpoint, and $\text{cross_rate}(N)$ is the S2n cross-property fracture rate. The scoring function uses equal weights for $\log_{10}(F)$ and cross_rate , which is a currently reasonable operational choice (the subjectivity of scoring function weight selection and multi-scheme sensitivity analysis results are discussed in §4.4 Candid Acknowledgment).

The full N-axis scoring curve detected 17 local peaks: N=19, 23, 35, 42, 54, 60, 64, 74, 81, 86, 93, 99, 111, 117, 123, 130, 137. The scoring rankings of confirmed breakpoints on the full N-axis are shown in Table 8.

Table 8: Scoring Rankings of Confirmed Breakpoints on the Full N-Axis

Confirmed Bkp	$\log_{10}(F)$	cross_rate	Score	Rank/Total	Percentile	Status
N=23	3.27	0.846	5.67	6/135	Top 4.4%	✓ Extremely Significant
N=40	3.29	0.556	5.00	33/135	Top 24.4%	✓ Significant
N=64	3.04	0.667	5.01	32/135	Top 23.7%	✓ Significant
N=129	2.29	1.000	4.25	62/135	Top 45.9%	⚠ Marginal
N=108	2.51	0.313	3.16	100/135	Top 74.1%	Weaker Signal
N=89	2.72	0.091	2.94	114/135	Top 84.4%	Interaction-Dominant
N=145	2.04	0.125	2.21	134/135	Top 99.3%	Insufficient Sample

N=23, N=40, and N=64 all have scores within the top 25%—they are naturally prominent peaks on the full N-axis scoring curve, not points cherry-picked from noise. All 135 candidate positions underwent the same scoring process, and readers can see the complete evidence distribution. This directly addresses the "post-hoc discovery" criticism.

N=89 and N=108 rank lower. The lower ranking of N=89 is expected—its function in the binding energy equation is mainly realized through interaction effects ($\eta=1.61$) rather than direct structural fracture, playing the role of a "multi-clamp node" in the confirmed breakpoint network. The confirmed breakpoint status of N=89 was established through rigorous multi-tool screening in Papers 22–23 (one of the Bai-Perron BIC optimal breakpoints, passing multi-tool cross-validation); its low score reflects functional differentiation (interaction-effect-dominant rather than direct structural fracture), not instability of its statistical status. N=108 is located in the heavy nuclear region with relatively fewer samples, and its local Chow F is not as high as in the light nuclear region, but its cross-property verification still passes. N=145 ranks last for a clear reason—insufficient sample size in the superheavy nuclear region, with inherently low Chow F (109.64), but it still maintains signals in cross-property verification (S2n: 12.5%, $Q\alpha$: 40.0%).

The scoring rankings completely present the signal gradient within the confirmed breakpoint network—N=23 is the strongest, N=40/64 follow, and N=89/108/129/145 rank lower for their respective reasons. This gradient itself is valuable scientific information, reflecting the intensity differences of regime switching across different mass regions.

Table 9: Raw Data of Full N-Axis Local Peaks (Score > 4.0)

N	$\log_{10}(F)$	cross_rate	Score
19	3.27	0.857	6.07
23	3.27	0.846	5.67
35	3.00	0.750	5.25
42	2.96	0.714	5.07
54	2.72	0.889	5.14
60	2.82	0.833	5.17
64	3.04	0.667	5.01
74	2.62	0.750	4.59
81	2.49	0.818	4.53
86	2.39	0.833	4.38
93	2.38	0.750	4.17
99	2.47	0.667	4.12
111	2.38	0.750	4.17
117	2.39	0.714	4.10
123	2.18	0.833	4.00
130	2.22	0.667	3.70
137	2.08	0.750	3.64

To further test whether N=40 and N=64 should be elevated to confirmed breakpoints, this paper performed multi-tool cross-validation and cross-property verification.

Table 10: Independent Tool Cross-Validation for N=40 and N=64

Test Tool	N=40 Result	N=64 Result
Chow F	1941.62 (p=0.0000)	1097.41 (p=0.0000)
Variance Ratio (low/high)	0.07	0.09
inv Transform	F=12.33, p=0.0000	F=5.28, p=0.0051
log Transform	F=353.30, p=0.0000	F=216.00, p=0.0000
sqrt Transform	F=1434.68, p=0.0000	F=821.05, p=0.0000
Bai-Perron Local Breakpoints	31, 34, 44, 48	55, 56, 68, 72
S2n Fracture Rate	55.6% (10/18)	66.7% (14/21)
Q α Fracture Rate	40.0% (8/20)	23.8% (5/21)

N=40 and N=64 satisfy all criteria for confirmed breakpoints. This paper formally elevates them to confirmed breakpoints—the confirmed breakpoint network expands from 5 to 7.

BIC Audit: Model Selection Analysis with Forced Inclusion of N=40/64. To address the "BIC suppression" criticism, this paper performed a BIC audit with forced inclusion of N=40/64. The original 10-

breakpoint model has $BIC=38967.6$. After forced inclusion of $N=40$, BIC increases by $+129.1$; forced inclusion of $N=64$ increases BIC by $+129.7$; simultaneous inclusion of both increases BIC by $+135.1$. BIC indeed increases—because BIC favors more parsimonious models, using $N=89$ to cover the vast region from $N=28$ to $N=82$ at once is more "economical" than splitting at $N=40$ and $N=64$ twice. However, the economy of BIC does not equal physical reality—the local Chow F , multi-tool cross-validation, and cross-property fracture rates of $N=40$ and $N=64$ independently confirm them as genuine physical signals. The difference in absolute BIC values originates from the different data subsets used in this paper and Tang (2026v)[25] (this paper's scoring curve data contains 3558 nuclides, while Tang (2026v)[25] used 1368 complete data nuclides, with sample size increased by about 2.6 times, BIC absolute value scaled proportionally), and does not affect the interpretation of ΔBIC .

The traditional shell model interprets $N=40$ as a "subshell closure" ($2p_{1/2}$ orbital filled, weak shell effect), but statistical evidence indicates it is a major shell closure level—the Chow F at $N=40$ is 1941.62, comparable to the Chow F of 1891.10 at $N=28$. The traditional shell model interprets $N=64$ as a "subshell closure" ($1g_{9/2}$ orbital filled), but statistical evidence similarly indicates it is a major shell closure level—the Chow F at $N=64$ is 1097.41, comparable to the major shell closures between $N=50$ – $N=82$. The nuclear shell regime-switching theory reclassifies the structures of the traditional framework—this is a direct manifestation of the theory's discovery capability.

After expanding the confirmed breakpoint network from 5 to 7, Theorem 7 still holds—all six shell region deviations defined by confirmed breakpoints are zero.

Table 11: Extended Verification of Theorem 7 (7 Confirmed Breakpoints)

Shell Region	Interval	Internal Bkps	Cross Width	Internal Width Sum	Deviation
Light Cascade Region	20–23	[23]	3	0	0
Light-Medium Transition	23–40	[23, 40]	17	17	0
Medium Core Region	40–64	[40, 64]	24	24	0
Medium-Heavy Transition	64–89	[64, 89]	25	25	0
Heavy Boundary Region	89–108	[89, 108]	19	19	0
Heavy Stable Region	108–129	[108, 129]	21	21	0
Superheavy Region	129–145	[129, 145]	16	16	0

Note: Values in parentheses are shell region boundary positions defined by confirmed breakpoints. The Light Cascade Region (20–23) contains only one internal breakpoint $N=23$, with internal width sum=0, and the deviation is directly taken from the difference between cross width and internal width sum. Theorem 7 is the internal self-consistency condition of the confirmed breakpoint network (algebraic necessity); its scientific value lies in the directional advantage demonstrated in cross-network comparison (see §4.3 for details).

Extended version of Theorem 8 (Confirmed Breakpoint Cascade Skeleton Theorem): N=20 and N=28 share N=23 as a cascade node (spacing=0), and precise cascade is established. N=40 is added as a secondary cascade node between N=28→N=50, and N=64 is added as a secondary cascade node between N=50→N=82. N=89 remains the multi-clamp node for the three traditional magic numbers N=28, N=50, and N=82. The cascade skeleton expands from a single precise cascade pair to a multi-level cascade network.

Table 12: Summary of Core Statistics for Confirmed Breakpoints

Confirmed Bkp N	Chow F	Variance Ratio	S2n Fracture	Qα Fracture	CUSUM p-value	Permutation p
23	1879.47	0.02	84.6%	92.3%	<0.0001	0.0000
40	1941.62	0.07	55.6%	40.0%	<0.0001	0.0000
64	1097.41	0.09	66.7%	23.8%	0.0028	0.0000
89	525.87	0.22	9.1%	30.4%	<0.0001	0.0000
108	319.85	0.15	31.3%	17.6%	<0.0001	0.0000
129	195.25	0.18	100.0%	100.0%	<0.0001	0.0000
145	109.64	0.11	12.5%	40.0%	<0.0001	0.0000

After expanding the confirmed breakpoint network from 5 to 7, nuclides near the cascade node N=23 exhibit unique statistical characteristics. The sphericity gradient in the N=23 cascade region (N=20–28) is significantly higher than in non-cascade regions (mean 0.1947 vs 0.1362, p=0.0000). The nuclides with the most dramatic sphericity changes are shown in Table 13.

Table 13: Shape Coexistence Candidates (Top 10 Sphericity Gradients) at Cascade Node N=23

Nuclide	A	Z	N	Sphericity Gradient
⁵⁰ Ni	50	28	22	0.669
⁴⁸ Ni	48	28	20	0.666
⁵¹ Ni	51	28	23	0.643
⁴⁹ Ni	49	28	21	0.641
⁴⁷ Co	47	27	20	0.590
⁴⁶ Fe	46	26	20	0.573
⁵² Cu	52	29	23	0.572
⁵³ Cu	53	29	24	0.500
²⁸ O	28	8	20	0.473
⁵⁴ Zn	54	30	24	0.400

The nuclides with the most dramatic sphericity changes near the cascade node N=23 are all nickel isotope chains (⁴⁸Ni to ⁵¹Ni), along with neighboring isotopes of cobalt, iron, copper, and zinc. The nickel isotope chain is a known shape coexistence hotspot region in nuclear physics—⁵⁶Ni is a doubly magic nucleus (N=28,Z=28),

while ^{48}Ni ($N=20, Z=28$) is a doubly magic candidate awaiting experimental confirmation. The algorithm, completely unaware of this nuclear physics knowledge, purely data-drivenly flagged the nickel isotope chain as a "shape coexistence candidate"—this is independent verification of Theorem 8 (Confirmed Breakpoint Cascade Skeleton Theorem) of Tang (2026w)[27].

To test the robustness of the confirmed breakpoint network to data truncation, this paper performed a local stability test—re-examining the local Chow signals of all confirmed breakpoints after removing $N>150$. Results are shown in Table 14.

Table 14: Local Stability Test (After Removing $N>150$)

Confirmed Bkp	Chow F	p-value	Status
N=23	2306.16	1.11×10^{-16}	✓
N=40	1630.53	1.11×10^{-16}	✓
N=64	883.32	1.11×10^{-16}	✓
N=89	406.68	1.11×10^{-16}	✓
N=108	216.95	1.11×10^{-16}	✓
N=129	98.88	1.11×10^{-16}	✓

After removing $N>150$, all confirmed breakpoints remain extremely significant—the core network structure is robust to data truncation, and the confirmed breakpoint network is not an accidental product of the full dataset.

4. Discussion

4.1 Cross-Property Universality as Statistical Falsification of the Shell Model Axiom

The cross-property universality of the confirmed breakpoint network constitutes a direct statistical falsification of the axiom that "nucleons move independently in a static mean field". If independent particle motion were the basic organizing principle, structural fractures in the binding energy equation should not consistently appear in separation energies and decay energies—because these physical quantities involve transitions between different energy states, and their statistical behavior is not entirely determined by ground-state single-particle energy levels. The dual 100% fracture rate of $S2n$ and $Q\alpha$ at $N=129$ means that the regime boundary has exactly the same position across three independent physical quantities—this cannot be explained by the single-particle picture.

This finding, together with Theorem E (Incompleteness Theorem of the Independent Particle Motion Axiom) of Tang (2026x)[28], constitutes a complete multi-dimensional verification: Theorem E proves the incompleteness of independent particle motion as a many-body system statistical description framework from

the theoretical derivation level, while this paper provides direct statistical falsification from the cross-property empirical testing level.

4.2 Two-Dimensional Structure as an Independent Challenge to the Spherically Symmetric Mean Field

The mean field of the shell model is spherically symmetric—proton and neutron single-particle energy levels are independent of each other, and no N-Z synergistic interaction effects exist. However, this paper finds that the interaction terms of Z with all three execution factors are highly significant, and the $N \times Z$ synergistic effect confirms that neutron-proton asymmetry is an independent regime modulator. This means that regime switching has coupling in both proton and neutron dimensions—the spherically symmetric mean field cannot describe this coupling.

The two-dimensional structure of the confirmed breakpoint network and the independent parallel modulation of color charge and flavor quantum numbers as orthogonal meta-regime factors in particle physics (Tang, 2026n-q)[18-21] constitute cross-disciplinary isomorphic verification—in different disciplines, the orthogonal independence of regime factors is manifested as two-dimensional coupling of regime switching. Five disciplines, one core logic: the "basic concepts" in traditional theories are projections of regime switching under specific limits, while the regime boundaries revealed by the Factor Hierarchy Law are the genuine fundamental quantities in finite systems.

4.3 Refinement of the Confirmed Breakpoint Network and the Discovery Capability of the Nuclear Shell Regime-Switching Theory

The elevation of N=40 and N=64 demonstrates the discovery capability of the nuclear shell regime-switching theory. The traditional shell model interprets N=40 and N=64 as "subshell closures" (weak shell effects), but the screening criteria of the nuclear shell regime-switching theory (multi-tool cross-validation + cross-property verification + full N-axis scoring curve) correctly position them as major shell closure-level confirmed breakpoints. The full N-axis scoring curve confirms that they are naturally prominent peaks in the complete evidence distribution, not points cherry-picked from noise—this directly addresses the "post-hoc discovery" criticism. After forced inclusion of N=40/64, BIC increases—the economy of BIC does not equal physical reality, and the local Chow F and cross-property fracture rates independently confirm them as genuine physical signals.

Theorem 7 is the internal self-consistency condition of the confirmed breakpoint network (algebraic necessity). Its scientific value lies in the directional advantage demonstrated in cross-network comparison—the deviation of the confirmed breakpoint network under traditional magic number shell regions (26.5) is numerically superior to the deviation of the traditional magic number network under confirmed breakpoint shell

regions (32.7). It should be candidly acknowledged that this numerical advantage has not reached traditional statistical significance thresholds under the current sample size ($p=0.77$ for strict threshold $F>100$, $p=0.75$ for relaxed threshold $F>50$). Therefore, the cross-network comparison value of Theorem 7 should currently be regarded as a directional, numerical advantage rather than decisive statistical verification.

Table 15: Discovery Capability Comparison Between the Nuclear Shell Regime-Switching Theory and the Shell Model

Structure	Predictable by Shell Model?	Predictable by Regime-Switching Theory?	Verification Status
Existence of confirmed breakpoint network	No	Yes	Verified by Tang (2026v)[25]
N=129 cross-property dual 100% fracture	No	Yes	Verified in this paper
Two-dimensional (N,Z) structure of confirmed breakpoint network	No	Yes (guided by theoretical logic)	Discovered in this paper
N=40 and N=64 as major shell closure levels	No (underestimated as subshell)	Yes	Discovered in this paper

Theorem 8 expands from a single precise cascade pair to a multi-level cascade network. The nickel isotope chain near the cascade node N=23 is purely data-drivenly flagged as a shape coexistence candidate, consistent with known experiments—this is independent verification of Theorem 8.

4.4 Candid Acknowledgment

All empirical findings of this paper are based on a series of preprint studies by the same author (Tang, 2026v–y)[25-28] and have not yet undergone independent third-party replication. Theorem 7 is the internal self-consistency condition of the confirmed breakpoint network (algebraic necessity), and its cross-network comparison value currently manifests as a directional advantage (numerically superior to the traditional magic number network, but not reaching statistical significance). The cross-data environment replication of Theorem C was completed between different code implementations by the same author and does not constitute independent replication in a strict sense. N=40 and N=64 are positioned as "secondary confirmed breakpoints"—their physical reality requires further confirmation through future experimental data and multi-tool cross-validation. The positioning of N=89 as "interaction-effect-dominant" currently relies on its η value (1.61) and Theorem 8, and future quantitative verification through Z-axis cross-property testing is needed. The intercept/slope ratio includes interpolated and extrapolated values, which have been explicitly distinguished in the data flags. Missing value sensitivity analysis confirms that after randomly deleting 10% of data, Chow F changes by only 10.2%, and core conclusions are robust within reasonable missing ranges.

The scoring function uses equal weights for $\log_{10}(F)$ and cross_rate , which is a currently reasonable operational choice. Multi-scheme sensitivity analysis (Table 16) shows that $N=40$ and $N=64$ exceed the 30% threshold in rankings when only considering structural fracture intensity or only cross-property universality, but both remain within the top 40% across all schemes, significantly better than random expectation (50%). Under the composite scheme (multiplicative form), $N=40$ ranks in the top 25.2%, and $N=64$ ranks in the top 24.4%. The elevation of both is based on comprehensive evidence from multi-tool cross-validation (Table 10), cross-property verification (Table 4), and permutation testing (Table 12); the scoring ranking is an auxiliary indicator, not the sole criterion.

Table 16: Multi-Scheme Sensitivity Analysis of Scoring Functions

Scheme	N=23	N=40	N=64	N=89	N=108	N=129	N=145
v1: $\log F$ only	5.9%	17.0%	37.0%	55.6%	69.6%	85.2%	97.0%
v2: cross_rate only	18.5%	40.0%	25.9%	92.6%	74.8%	8.1%	90.4%
v3: additive	3.7%	18.5%	25.2%	75.6%	79.3%	57.8%	100.0%
v4: multiplicative (orig.)	5.9%	25.2%	24.4%	85.2%	74.8%	46.7%	100.0%

Note: Table values are the percentile rankings (top X%) of each confirmed breakpoint among 135 candidate positions. $N=40$ ranks in the top 40.0% under v2, and $N=64$ ranks in the top 37.0% under v1, both marginal exceedances. Under the composite scheme (v4), both rank around the top 25%.

5. Conclusions

This paper conducted a direct empirical test of the core predictions of the nuclear shell regime-switching theory and demonstrated the theory's discovery capability. The main conclusions include:

(1) The confirmed breakpoint network consistently passes cross-property verification on two independent physical quantities—two-neutron separation energy and α -decay energy—with both quantities achieving a 100% fracture rate at $N=129$, and both exceeding 84% at $N=23$. The confirmed breakpoint network is not a statistical artifact of the binding energy equation but a genuine many-body quantum system structure. The shell model cannot explain this discovery.

(2) For the first time, the confirmed breakpoint network is found to possess a rigorous two-dimensional (N, Z) structure— Z itself satisfies all defining characteristics of a regime factor, and the N - Z synergistic effect confirms that neutron-proton asymmetry is an independent regime modulator. Regime switching has coupling in both proton and neutron dimensions—the spherically symmetric mean field of the shell model cannot describe this coupling. This is the first purely data-driven two-dimensional regime-switching network in the history of nuclear physics.

(3) $N=40$ and $N=64$ are identified as confirmed breakpoints for the first time—integrating multi-dimensional evidence, both rank stably within the top 40% across multi-scheme scoring, and within the top 25% under the composite scheme. The traditional shell model underestimates them as "subshell closures", while the nuclear shell regime-switching theory correctly positions them as major shell closure levels. The full N -axis scoring curve confirms that they are naturally prominent peaks in the complete evidence distribution, not points cherry-picked from noise. After forced inclusion of $N=40/64$, BIC increases—the economy of BIC does not equal physical reality. After expanding the confirmed breakpoint network from 5 to 7, the deviations of Theorem 7 across six shell regions are all zero, and the extended version of Theorem 8 adds secondary cascade nodes. The confirmed breakpoint network remains robust after removing the superheavy region.

This paper releases version 1.0 of the Regime Nuclide Map and the nuclide regime coordinate database (3558 nuclides $\times$ 25 fields) as open data assets. Part II (Tang, 2026ab) will enter deeper statistical dimensions—physical localization of shell correction energy, verification of Theorem 10 (Multidimensional Response Theorem), cross-data environment replication of Theorem C, and methodological contributions of information duality.

Data Availability Statement

The Regime Nuclide Map (zonal and continuous versions) and the nuclide regime coordinate database (regime_nuclide_chart.csv, 3558 nuclides $\times$ 25 fields) generated in this paper are uploaded to ZENODO along with this paper. Basic nuclear data comes from the AME2020 and NUBASE2020 public databases (Wang et al., 2021)[29], comprising six independent data files. The confirmed breakpoint network has been verified across six independent data dimensions, including cross-regime fracture testing of the covariance matrix. All analysis code is available upon request.

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Appendix A: Field Descriptions of the Nuclide Regime Coordinate Database

Field Name	Type	Description
A	int	Mass number
Z	int	Proton number
N	int	Neutron number
element	str	Element symbol
bind_per_A_keV	float	Binding energy per nucleon (keV)
shell_correction	float	Shell correction energy (keV)
paper_zone	str	Nuclear zone classification (Light/Medium/Heavy)
n_regime	str	N-regime classification
z_regime	str	Z-regime classification
regime_id	str	Regime ID (n_regime_z_regime)
nearest_bkp_type	str	Nearest breakpoint type (confirmed/candidate/magic)
boundary_proximity	str	Boundary proximity (adjacent/near/interior)
dist_to_magic_N	int	Distance to nearest neutron magic number
dist_to_magic_Z	int	Distance to nearest proton magic number
dist_to_confirmed_bkp	int	Distance to nearest confirmed breakpoint
dist_to_any_bkp	int	Distance to nearest blind breakpoint
sphericity_weight_confirmed	float	Sphericity weight based on confirmed breakpoints
sphericity_weight_magic	float	Sphericity weight based on traditional magic numbers
interpolated_ratio	float	Intercept/slope ratio (interpolated)
ratio_data_flag	str	Ratio data flag (measured/interpolated/extrapolated)
near_cross_point	int	Near dual-track crossing point (0/1)
in_silence_zone	int	In mid-shell silence zone (0/1)
is_double_magic	int	Is doubly magic (0/1)
N23_cascade	int	In N=23 cascade region (0/1)
N89_multi_clamp	int	In N=89 multi-clamp region (0/1)